IndiCom Preregistration:

Title: The Heterogeneity of Common Ground: Recruitment of ConContext Information for the Interpretation of Indirect Answers (to Polar Questions)

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Description: When someone asks a polar question such as “Will you be at the beach tomorrow?”, an indirect answer like “It’s going to be windy” is ambiguous and requires contextual information about the speaker to be disambiguated. If it is common ground that the speaker is a surfer, the likely interpretation is “yes”; if the speaker only goes to the beach to tan and relax in the sun, it is likely a “no”. Indirect answers to polar questions thus serve as a convenient testing ground for exploring the heterogeneity of information in the common ground: whether contents that enter the common ground with different information statuses (e.g. at-issue vs. not-at-issue, explicit vs. implicit) are represented differently and differ in their accessibility for recruitment for downstream tasks like disambiguation, or whether or information is represented the same and just as accessible.

Research Question & Hypotheses:

We explore whether there is a heterogeneity of information within the common ground: whether content that has entered the common ground with different information statuses (at-issue vs. not-at-issue) or different explicitness (explicit vs. implicit) is represented differently and thus differently accessible and available for recruitment for downstream tasks. Since there is some assumption that clausal not-at-issue constructions (e.g. NRRCs) may be at-issue at the time of processing (by having their own QUD that they are at-issue for), we further distinguish between potentially no-longer-at-issue content (clausal not-at-issue content) and never-at-issue content (e.g. presupposed content), exploring whether these nuances in not-at-issue content lead to differences in the common ground.

We formulate the following hypotheses:

(1) We hypothesize that at-issue content is more accessible and will thus lead to stronger disambiguation of indirect answers to polar questions than both the no-longer-at-issue condition as well as the never-at-issue conditions, which we expect to lead to a weaker disambiguation of the indirect answers.

(2) We hypothesize that explicit content is more accessible and will thus lead to stronger disambiguation of indirect answers to polar questions than the implicit content, which we expect to lead to a weaker disambiguation of the indirect answers.

(3) We hypothesize that no-longer-at-issue content is more accessible and will thus lead to stronger disambiguation of indirect answers to polar questions than the never-at-issue content, which we expect to lead to a weaker disambiguation of the indirect answers.

We thus expect the following ordinal relation of disambiguation strength:

at-issue content > no-longer-at-issue content > never-at-issue content

explicit content > implicit content

Design Plan:

Study Type: Experiment

Blinding: The relevant manipulation is within-participants. The experiment is conducted via the internet. No direct contact between experimenters and participants will take place.

Study Design: The study is a 2 (polarity) × 4 (informations status) factorial within-subjects web-based experiment, in which participants are asked to play through a small story game during which they read a small dialogue which sets up a common ground between them and a fictional interlocutor. This first dialogue introduces the relevant context information in one of the 4 different information statuses - at-issue explicit, at-issue implicit, no-longer-at-issue or never-at-issue - and one of the two different polarities - negative, positive. After this common ground is established, the participant is shown a dialogue where they ask a polar question and the fictional interlocutor answers with an indirect answer. The participant is asked to interpret the indirect answer on a 7-point scale, with 1 meaning “definitely no” and 7 meaning “definitely yes”. The participant is then given a forced choice task between 3 possible answer options on how they want to respond to the indirect question, which serves as a secondary control.

Randomization: Each participant completes 8 main trials, seeing each of the 8 conditions exactly once. For each main trial, the participant is given the choice between three possible items (conversation topics) that they can choose from. Once an item was chosen by the participant, it is removed from the options. The order in which the items are presented as options are entirely randomized.

Sampling Plan:

Existing Data: Data from a previous pilot study (N=15) was available and guided the specification of statistical models. This data is not included in the final analysis. No data from the experiment to be preregistered here was available at the time of preregistration.

Explanation of existing data: Existing data does not enter into future analysis.

Data collection procedures: Participants will be recruited on Prolific.com and compensated £2.25 per participant. Participants are required to have English as their native language. There are three control questions to ensure participants are reading the dialogues attentively and remembering the critical information correctly. If a participant fails a control question, they are reminded to read the dialogues more carefully.

Sample size: We will collect a total of 200 subjects and will analyze all subjects that meet inclusion criteria.

Sample size rationale: We performed a Bayesian assurance analysis, using the model fitted to our pilot data to simulate experiments at different sample sizes, analyzing those to record how often the target effect is recovered. As in the analysis plan, we will count as an effect those runs where the 95% credible interval excludes 0.

We simulated 250 runs per participant number, finding the following results:

for N = 50:

contrast	detection_rate	mean_effect	mean_effect_ci_lower	mean_effect_ci_upper	sd_across_simulations
atissue_vs_never	0.840	2.4218101	2.2736099	2.5670330	1.175585
explicit_vs_implicit	0.436	0.7549839	0.6244863	0.8816690	1.036551
nolonger_vs_never	0.280	0.4267516	0.2827287	0.5721972	1.147335

for N = 75:

contrast	detection_rate	mean_effect	mean_effect_ci_lower	mean_effect_ci_upper	sd_across_simulations
atissue_vs_never	0.916	2.4438312	2.2954882	2.5857730	1.172156
explicit_vs_implicit	0.548	0.7308374	0.6007703	0.8610021	1.071583
nolonger_vs_never	0.336	0.5561942	0.4300859	0.6867246	1.045769

for N = 100:

contrast	detection_rate	mean_effect	mean_effect_ci_lower	mean_effect_ci_upper	sd_across_simulations
atissue_vs_never	0.916	2.4616792	2.3091216	2.6086007	1.199450
explicit_vs_implicit	0.544	0.7633787	0.6323847	0.8915512	1.048926
nolonger_vs_never	0.432	0.4477185	0.3118245	0.5847337	1.111999

result 50 participants, 250 simulations:
at-issue vs never: 0.84

explicit vs implicit: 0.436
nolonger vs. never: 0.28

result 100 participants, 250 simulations:
at-issue vs never: 0.916
explicit vs implicit: 0.544
nolonger vs. never: 0.432

Stopping Rule: none

Variables:

Manipulated variables: We manipulate two independent variables:
Information status (4 levels): at-issue explicit, at-issue implicit, no-longer-at-issue, never-at-issue
Polarity (2 levels): positive, negative

Measured variables: We record the following three responses provided by the participants.

- **Interpretation rating**: Participants rate their interpretation of the indirect answer on a 7-point Likert scale ranging from 0 ("definitely no") to 7 ("definitely yes").
- **Forced-choice response**: Participants select one of three response options indicating how they would respond to the indirect answer.
- **Control question accuracy**: Responses to comprehension/control questions assessing whether participants correctly processed and remembered the critical contextual information.

Indices:

Analysis Plan:

Statistical Models:

We will fit a Bayesian ordinal regression model using the brms R package. The dependent variable is the interpretation rating (0–7 scale). The model will use a cumulative logit link function.
The model in brms syntax is:

rating ~ informationStatus * polarity + (1 + informationStatus * polarity | submission_id) + (1 | itemID)

Transformations:

What we are interested in are the differences between the polarities for each condition, so for each condition, we will compute the delta between negative and positive context:

Δ = positive − negative

This yields the following condition-specific polarity effects:

- Δ_atissue_explicit
- Δ_atissue_implicit
- Δ_nolonger_atissue
- Δ_never_atissue

To test whether there are significant differences between the different information statuses, we will then compare the different conditions to each other:

Δ_atissue_explicit − Δ_atissue_implicit
Δ_atissue − Δ_nolonger_atissue

Δ_nolonger_atissue − Δ_never_atissue

We will then report posterior means, 95% credible intervals and probability effects.

Inference Criteria:

Evidence for a directional effect will be considered stronger when the posterior distribution of the relevant contrast is consistently above or below 0.

Data Exclusion:

We will exclude data from participants that fail two or more of the three control questions, suggesting they did not read the experiment carefully.

Missing data:

If data isn't recorded for some but not all of the trials for a participant, we will include the trials from that participant for which we have data.